

1.5.8 Power Series

The mathematical operations we have discussed so far are *exact*. Given precise input, their results are exact formulae.

In many situations, however, you do not need an exact result. It may be quite sufficient, for example, to find an *approximate* formula that is valid, say, when the quantity x is small.

This gives a power series approximation to $(1+x)^n$ for x close to 0, up to terms of order x^3 .

```
In[1]:= Series[(1 + x)^n, {x, 0, 3}]
Out[1]= 1 + n x + ( $\frac{-n}{2} + \frac{n}{2}$ ) x2 + ( $\frac{n}{3} - \frac{n}{2} + \frac{n}{6}$ ) x3 + 0[x]4
```

Here is an approximation for x close to 1.

```
In[2]:= Series[(1 + x)^n, {x, 1, 2}]
Out[2]= 2n +  $\frac{2^n n (-1 + x)}{2}$  + 2n ( $\frac{-n}{8} + \frac{n}{8}$ ) (-1 + x)2 + 0[-1 + x]3
```

Mathematica knows the power series expansions for many mathematical functions.

```
In[3]:= Series[Exp[-a t] (1 + Sin[2 t]), {t, 0, 4}]
Out[3]= 1 + (2 - a) t + (-2 a +  $\frac{a^2}{2}$ ) t2 + ( $-\frac{4}{3} + a^2 - \frac{a^3}{6}$ ) t3 + ( $\frac{4a}{3} - \frac{a^3}{3} + \frac{a^4}{24}$ ) t4 + 0[t]5
```

If you give it a function that it does not know, Series writes out the power series in terms of derivatives.

```
In[4]:= Series[1 + f[t], {t, 0, 3}]
Out[4]= (1 + f[0]) + f'[0] t +  $\frac{f''[0] t^2}{2}$  +  $\frac{f^{(3)}[0] t^3}{6}$  + 0[t]4
```

Here is a simple power series, accurate to order x^5 .

```
In[5]:= Series[Exp[x], {x, 0, 5}]
Out[5]= 1 + x +  $\frac{x^2}{2}$  +  $\frac{x^3}{6}$  +  $\frac{x^4}{24}$  +  $\frac{x^5}{120}$  + 0[x]6
```

When you do operations on a power series, the result is computed only to the appropriate order in x .

```
In[6]:= %^2 (1 + %)
Out[6]= 2 + 5 x +  $\frac{13 x^2}{2}$  +  $\frac{35 x^3}{6}$  +  $\frac{97 x^4}{24}$  +  $\frac{55 x^5}{24}$  + 0[x]6
```

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This turns the power series back into an ordinary expression.

```
In[7]:= Normal[%]
```

$$\text{Out}[7]= 2 + 5 x + \frac{13 x^2}{2} + \frac{35 x^3}{6} + \frac{97 x^4}{24} + \frac{55 x^5}{24}$$

Now the square is computed *exactly*.

```
In[8]:= %^2
```

$$\text{Out}[8]= (2 + 5 x + \frac{13 x^2}{2} + \frac{35 x^3}{6} + \frac{97 x^4}{24} + \frac{55 x^5}{24})^2$$

Applying `Expand` gives a result with ten terms.

```
In[9]:= Expand[%]
```

$$\begin{aligned} \text{Out}[9]= & 4 + 20 x + 51 x^2 + \frac{265 x^3}{3} + \frac{467 x^4}{4} + \frac{1505 x^5}{12} + \\ & \frac{7883 x^6}{72} + \frac{1385 x^7}{18} + \frac{24809 x^8}{576} + \frac{5335 x^9}{288} + \frac{3025 x^{10}}{576} \end{aligned}$$

`Series[expr, {x, x0, n}]`

find the power series expansion of *expr* about the point $x = x_0$ to at most *n* terms

`Normal[series]`

truncate a power series to give an ordinary expression

Power series operations.